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1 Introduction

The rapid advancement of digital technologies has significantly transformed the way businesses operate, particularly in the hospitality industry. Restaurants are increasingly adopting online systems to manage their operations efficiently and enhance customer experience. This report presents the Object-Oriented Analysis and Design of an online restaurant management system developed for Gokyo Bistro, a popular dining establishment located in Jhamsikhel, Lalitpur.

Gokyo Bistro has been experiencing challenges in managing table reservations, handling increasing customer demand, and organizing orders effectively. Additionally, the growing need for home delivery services and customer engagement has highlighted the limitations of the existing manual system. To address these issues, a comprehensive online system is proposed to automate and streamline core restaurant operations.

The proposed system includes functionalities such as user registration and login, table reservation, beverage selection, online payment, food ordering with home delivery options, bill generation, menu management, reporting, and customer feedback. The system also provides administrative features such as generating financial reports, updating menu items, and managing promotional offers.

This report focuses on applying Object-Oriented Analysis and Design (OOAD) principles to model the system. Various UML diagrams, including Use Case Diagrams, Class Diagrams, Sequence Diagrams, and Activity Diagrams, are used to represent system behavior and structure. Additionally, project planning techniques such as Work Breakdown Structure (WBS) and Gantt Chart are utilized to outline the development process.

The objective of this report is to provide a clear and structured design of the proposed system, ensuring scalability, maintainability, and efficiency. The design serves as a foundation for future implementation and deployment of the system.

2 WBS and Gantt Chart

Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) for the Gokyo Bistro Management System provides a hierarchical decomposition of the entire project into smaller, manageable components. It begins with the overall project at the top level and is further divided into major phases such as project initiation, requirement analysis, system design, development, testing, deployment, and maintenance. Each of these phases is then broken down into specific tasks and sub-tasks, enabling a clear understanding of the scope of work. The WBS helps in organizing the project systematically, assigning responsibilities, and ensuring that all required activities are identified and completed efficiently.

The WBS is organized into **6 phases** matching Agile-Scrum sprints:

- **Sprint 0** — Project initiation: requirements, backlog, architecture, team setup
- **Sprint 1** — Login/Registration + Table Booking (core features first)
- **Sprint 2** — Beverage choices, advance payment (Rs.1200), cancellation logic (Rs.500 fee)
- **Sprint 3** — Order placement, home delivery, menu management
- **Sprint 4** — Billing, financial reports, revenue prediction, offers/discounts
- **Sprint 5** — Ratings & feedback, integration testing, UAT, final deployment

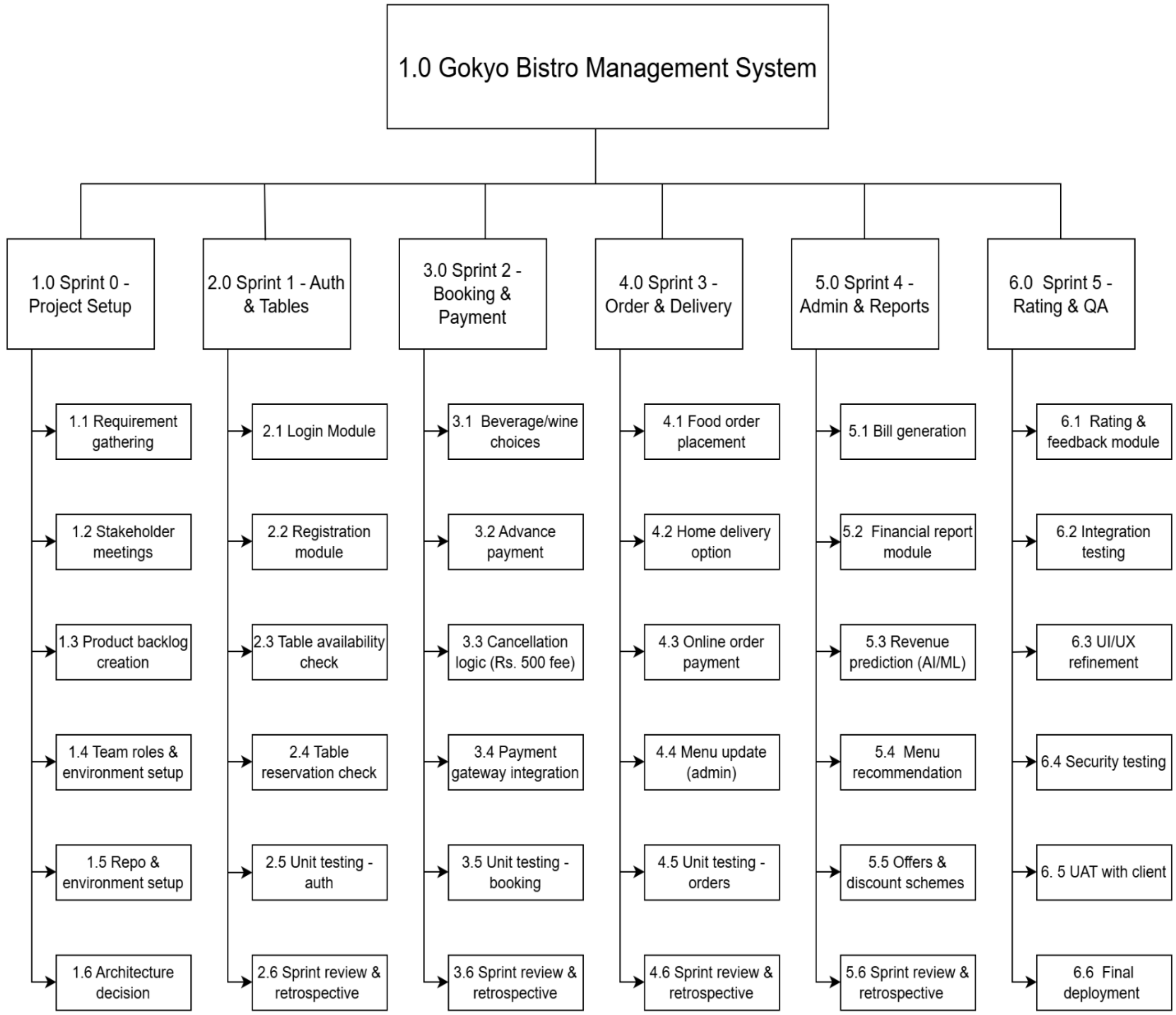

Figure 1: Work Breakdown Structure

Gantt Chart

The Gantt Chart for the project illustrates the timeline and scheduling of all activities involved in the development of the Gokyo Bistro Management System. It provides a visual representation of tasks along with their start and end times, showing the sequence and duration of each phase. Based on the Agile (Scrum) methodology, the chart organizes tasks into iterative stages, allowing flexibility and continuous progress throughout the development cycle. The Gantt Chart helps in tracking project progress, managing time effectively, and ensuring that all tasks are completed within the planned schedule.

The timeline spans **14 weeks** total:

- 6 sprints × 2 weeks = 12 weeks of active development
- 2 weeks buffer for UAT and final polish

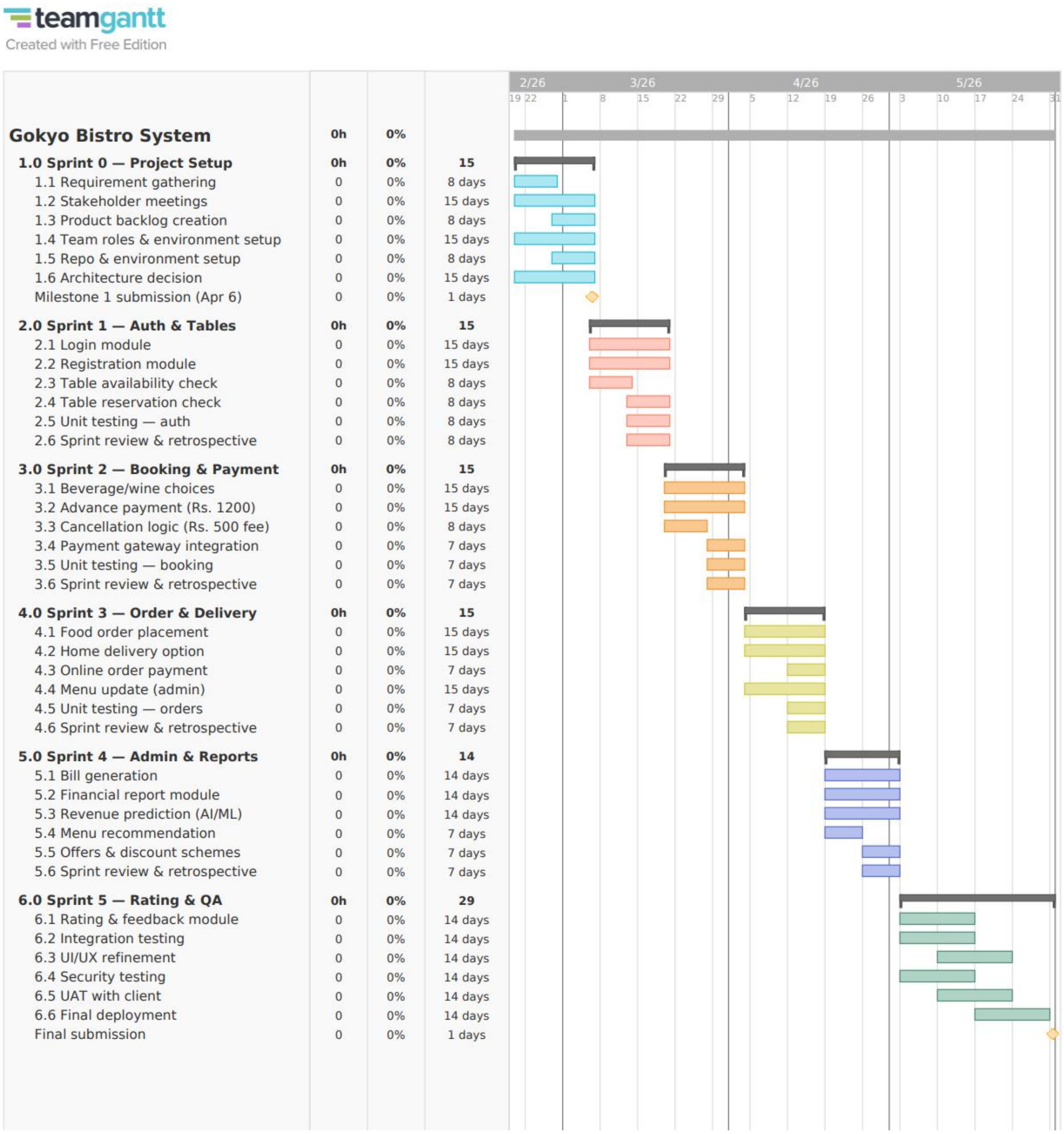

Figure 2: Gantt Chart

3 Use Case

The Use Case Model is an essential component of Object-Oriented Analysis and Design (OOAD) that helps in capturing the functional requirements of a system from the user's perspective. It identifies the interactions between different types of users (actors) and the system, illustrating how the system should behave in various scenarios. For the Gokyo Bistro Management System, the Use Case Model is developed to represent key functionalities such as user registration and login, table booking, payment processing, order placement, menu management, and feedback submission.

The model consists of a Use Case Diagram, High-Level Use Case Descriptions, and Expanded Use Case Descriptions. The Use Case Diagram visually represents the relationship between actors and system functionalities, providing an overview of system operations. High-Level Use Case Descriptions briefly explain each use case, outlining the main purpose and interactions. Furthermore, Expanded Use Case Descriptions are developed for selected use cases to provide detailed step-by-step flows, including preconditions, main flow, alternate flow, and postconditions. This structured approach ensures a clear understanding of system behavior and supports effective system design and development.

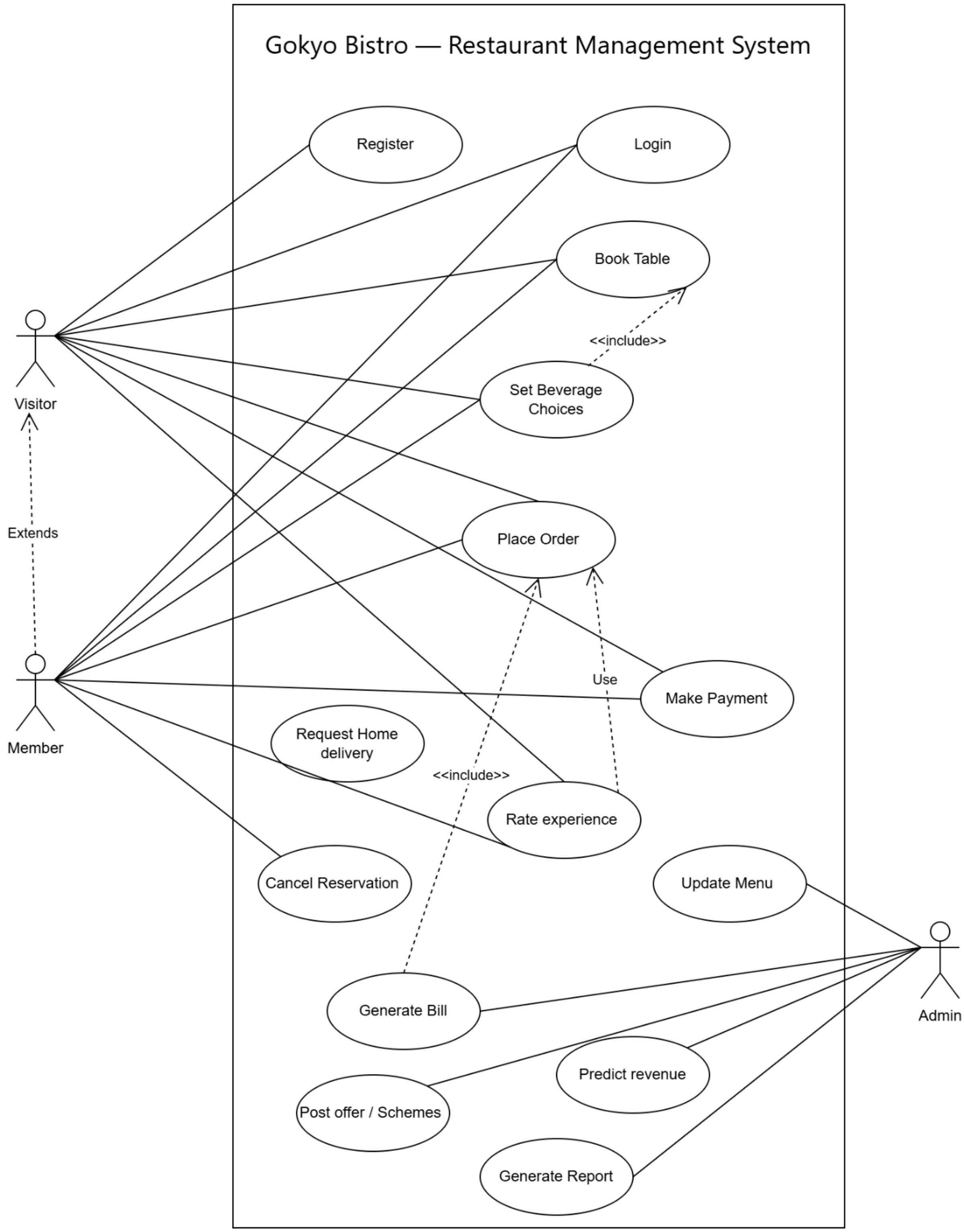

Figure 3: Use Case diagram

4 High Level Use Case

The High-Level Use Case Descriptions provide a brief overview of each functionality of the system by identifying the primary actors and their interactions with the system. These descriptions summarize the purpose of each use case without going into detailed steps. In the Gokyo Bistro Management System, high-level descriptions are used to outline key features such as booking a table, making payments, placing orders, managing menu items, and generating reports. This helps in understanding the overall system functionality in a simple and concise manner before moving into more detailed analysis.

UC1 — Login

- Use Case Name: Login
- Actor(s): Admin, Member
- Description: The user enters their credentials (email and password) to access their respective dashboard in the system.
- Pre-condition: The user must have a registered account in the system.
- Post-condition: The user is successfully authenticated and redirected to their role-based dashboard.

UC2 — Register

- Use Case Name: Register
- Actor(s): Visitor
- Description: A visitor creates a new member account by providing personal details such as name, email, phone number, and password.
- Pre-condition: The visitor must not have an existing account in the system.
- Post-condition: A new member profile is created and stored in the system database.

UC3 — Book Table

- Use Case Name: Book Table
- Actor(s): Visitor, Member

- Description: The user checks the availability of tables for a chosen date and time and reserves a table for the specified number of guests.
- Pre-condition: At least one table must be available for the requested date and time.
- Post-condition: The table is reserved, advance payment of Rs. 1200 is collected, and a booking confirmation is sent to the user.

UC4 — Set Beverage Choices

- Use Case Name: Set Beverage Choices
- Actor(s): Visitor, Member
- Description: While booking a table, the user optionally selects expected beverage preferences from the available list. Premium members can additionally reserve specific wine or whiskey choices.
- Pre-condition: A table booking must be in progress (extends UC3).
- Post-condition: The selected beverage preferences are saved along with the reservation record.

UC5 — Make Payment

- Use Case Name: Make Payment
- Actor(s): Visitor, Member, Payment Gateway
- Description: The user pays an advance amount of Rs. 1200 to confirm the table reservation through the integrated payment gateway.
- Pre-condition: A table must be selected and ready for booking.
- Post-condition: Payment is confirmed, the booking status is updated to "Confirmed", and a receipt is generated.

UC6 — Cancel Reservation

- Use Case Name: Cancel Reservation
- Actor(s): Visitor, Member

- Description: The user cancels a confirmed table reservation. If the cancellation is made on the same day as the reservation, a fee of Rs. 500 is deducted before the refund is processed.
- Pre-condition: A confirmed reservation must exist for the user.
- Post-condition: The reservation is cancelled, the table status is set to Available, and the refund (full or partial) is processed via the payment gateway.

UC7 — Rate Experience

- Use Case Name: Rate Experience
- Actor(s): Visitor, Member
- Description: The user rates their overall experience by providing scores for food quality, staff behaviour, and ambience, and can optionally submit written feedback.
- Pre-condition: The user must have completed a visit or placed an order.
- Post-condition: The rating and feedback are stored in the system and are visible to the admin.

UC8 — Place Order

- Use Case Name: Place Order
- Actor(s): Member
- Description: A logged-in member selects food and beverage items from the menu, chooses between dine-in or home delivery, and places an order. Payment can be made online.
- Pre-condition: The member must be logged in and the menu must be available in the system.
- Post-condition: The order is placed and forwarded to the kitchen, and the payment is processed.

UC9 — Request Home Delivery

- Use Case Name: Request Home Delivery
- Actor(s): Member
- Description: A member requests home delivery for a placed order by providing a valid delivery address. An estimated delivery time is provided by the system.
- Pre-condition: An order must have already been placed (UC8). A delivery address is required.
- Post-condition: A delivery request is created and an estimated delivery time is communicated to the member.

UC10 — Reserve Wine / Whiskey

- Use Case Name: Reserve Wine / Whiskey
- Actor(s): Member (Premium)
- Description: Premium members can reserve specific wine or whiskey choices from a curated list while booking a table.
- Pre-condition: The user must be a verified premium member and a table booking must be in progress.
- Post-condition: The selected wine or whiskey reservation is noted and held for the member's visit.

UC11 — Update Menu

- Use Case Name: Update Menu
- Actor(s): Admin
- Description: The admin can add new menu items, edit existing item details such as price and description, or delete items that are no longer available.
- Pre-condition: The admin must be logged in to the system.
- Post-condition: The menu is updated and the changes are immediately reflected to all users browsing the system.

UC12 — Generate Bill

- Use Case Name: Generate Bill
- Actor(s): Admin
- Description: The admin generates an itemised bill for a customer's completed order or visit, and provides a downloadable or printable receipt.
- Pre-condition: An order or reservation must exist and be in a completed state.
- Post-condition: A bill is generated and a receipt is made available for download or printing.

UC13 — Generate Report

- Use Case Name: Generate Report
- Actor(s): Admin
- Description: The admin generates financial reports showing total revenue, most popular menu items, booking trends, and other operational data for a specified period.
- Pre-condition: Sufficient transaction data must exist in the system.
- Post-condition: The report is generated and can be viewed on screen or exported.

UC14 — Post Offers / Schemes

- Use Case Name: Post Offers / Schemes
- Actor(s): Admin
- Description: The admin publishes special discount offers and promotional schemes from time to time, which are then made visible to visitors and members on the platform.
- Pre-condition: The admin must be logged in to the system.
- Post-condition: The offer is published and visible to all users on the platform until deactivated.

UC15 — Predict Revenue

- Use Case Name: Predict Revenue
- Actor(s): Admin
- Description: The system analyses historical order and revenue data to predict future revenue for upcoming years and provides recommendations on which menu items to promote.
- Pre-condition: Sufficient historical order and revenue data must be available in the system.
- Post-condition: Revenue forecasts and menu recommendations are displayed to the admin in the report dashboard.

5 Expanded Use Case

Expanded Use Case Descriptions provide a detailed and structured explanation of selected use cases by describing the complete interaction between the user and the system. These descriptions include important elements such as preconditions, main flow of events, alternate flows, and postconditions. In this project, two critical use cases are expanded to provide deeper insight into system behavior, ensuring that all possible scenarios are considered. This level of detail helps in improving system accuracy, reducing ambiguity, and serving as a strong foundation for further design and implementation phases.

Expanded Use Case — UC3: Book Table

- Use Case Name: Book Table
- Use Case ID: UC3
- Primary Actor: Visitor / Member
- Secondary Actor(s): Payment Gateway (via UC5 — Make Payment)
- Trigger: User clicks "Book a Table" on the homepage or navigation menu.
- Pre-conditions:
    - The system is online and accessible.

  - At least one table is available for the requested date and time.
- Post-conditions:
  - The table is reserved and its status is updated to "Reserved".
  - An advance payment of Rs. 1200 is successfully collected.
  - A booking confirmation is sent to the user via email or SMS.
- Include: Make Payment (UC5)
- Extend: Set Beverage Choices (UC4)

Main Flow (Basic Course of Events):

Table 1: Expanded Uses case (Book Table)

Step	Actor Action	System Response
1	User selects preferred date, time, and number of guests.	System queries the database and displays all available tables matching the criteria.
2	User selects a table from the list of available options.	System temporarily holds the selected table for 10 minutes to prevent double booking.
3	User optionally sets beverage or wine preferences (extends UC4).	System saves the beverage preferences with the reservation draft.
5	User confirms the booking and proceeds to payment.	System invokes Make Payment (UC5) to collect the Rs. 1200 advance.
6	Payment is successfully processed by the payment gateway.	System confirms the reservation, updates the table status to "Reserved", and sends a confirmation notification to the user.

Alternative Flow:

- Step 2a: If no tables are available for the selected date and time, the system displays a "No availability" message and suggests alternative available dates to the user.

Exception Flow:

- Step 4a: If the payment fails, the reservation draft is discarded, the table hold is released, and the user is notified of the payment failure with a prompt to try again.

Expanded Use Case — UC6: Cancel Reservation

- Use Case Name: Cancel Reservation
- Use Case ID: UC6
- Primary Actor: Visitor / Member
- Secondary Actor(s): Payment Gateway (for refund processing)
- Trigger: User clicks "Cancel Reservation" from their booking history page.
- Pre-conditions:
  - The user is logged in or has a valid booking reference number.
  - A confirmed reservation exists in the system for the user.
- Post-conditions:
  - The reservation status is updated to "Cancelled".
  - The reserved table status is set back to "Available".
  - A refund is processed: Rs. 1200 in full if cancelled before the reservation day, or Rs. 700 (Rs. 1200 minus Rs. 500 deduction) if cancelled on the same day.

Main Flow (Basic Course of Events):

Table 2: Expended Use Case (CancelReservation)

Step	Actor Action	System Response
1	User opens their booking history and selects the reservation they wish to cancel.	System fetches and displays the reservation details along with the applicable cancellation policy.
2	User clicks "Cancel" and confirms the cancellation on the confirmation prompt.	System checks whether the cancellation date matches the reservation date.

3		If the cancellation is on the same day: the system deducts Rs. 500 and sets the refund amount to Rs. 700. If before the reservation day: the system sets the full refund amount to Rs. 1200.
4		System initiates the refund transaction via the Payment Gateway.
5	User receives a cancellation confirmation notification showing the refund amount.	System updates the reservation status to "Cancelled" and the table status to "Available".

Alternative Flow:

- Step 1a: If the user does not have a registered account, they can still cancel using their booking reference number received at the time of booking.

Exception Flow:

- Step 4a: If the payment gateway fails to process the refund, the cancellation still proceeds and is recorded, but the refund is queued for manual processing. The user is notified that the refund may take additional time.

6 Sequence Diagram

A Sequence Diagram is a type of Unified Modeling Language (UML) diagram used to represent the dynamic behavior of a system by illustrating how objects interact with each other in a particular scenario over time. It focuses on the sequence of messages exchanged between different components or actors to accomplish a specific use case. In the context of the Gokyo Bistro Management System, the Sequence Diagram is used to model the step-by-step interaction involved in a selected use case, such as ordering food or booking a table.

The diagram visually represents objects, actors, and the messages passed between them in a time-ordered manner, typically from top to bottom. It helps in understanding how different parts of the system collaborate to perform a function, ensuring clarity in system behavior and logic flow. By modeling these interactions, the Sequence Diagram aids in identifying system responsibilities, improving design accuracy, and serving as a blueprint for implementation.

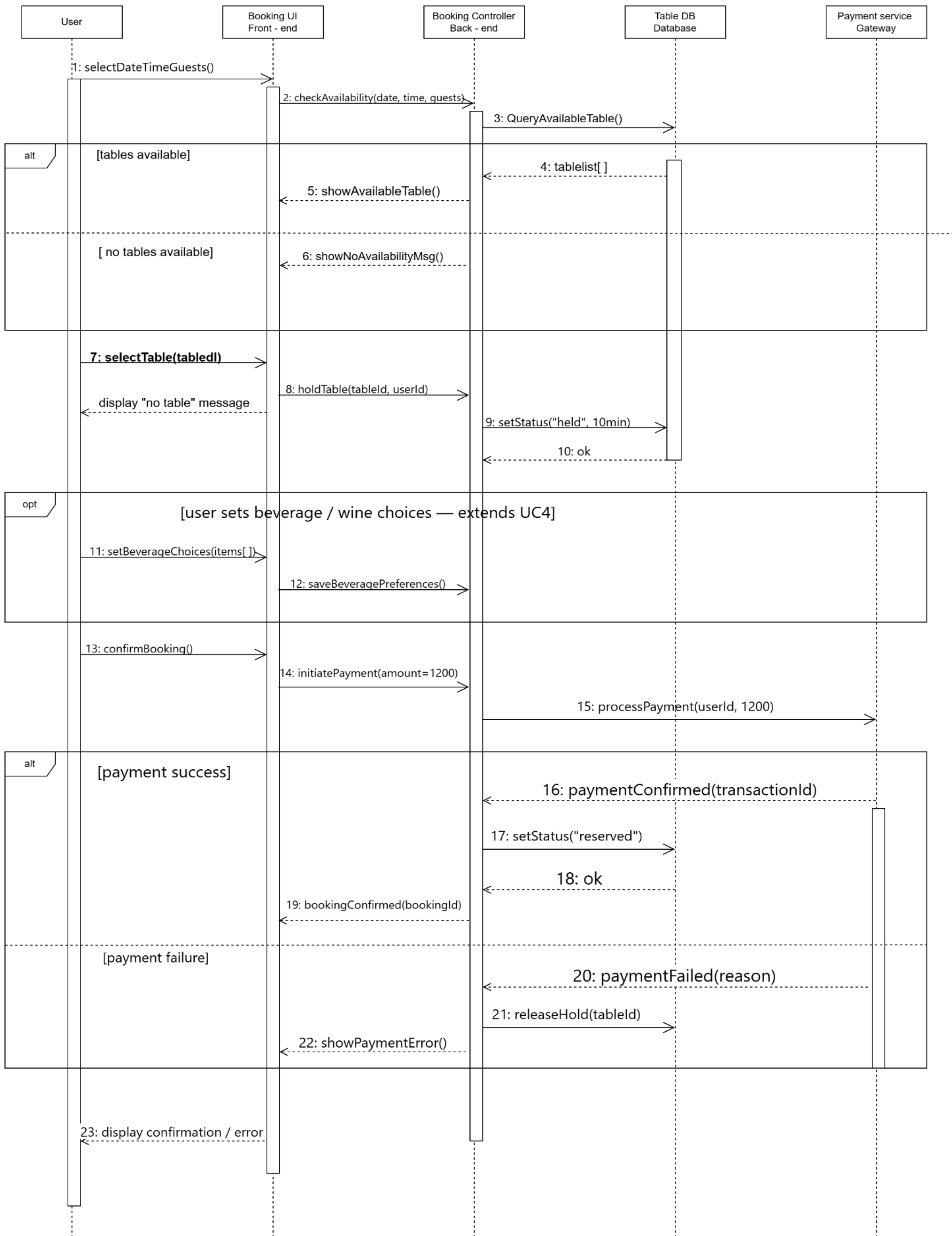

Figure 4: Sequence Diagramf

7 Activity Diagram

An Activity Diagram is a type of Unified Modeling Language (UML) behavioral diagram that represents the workflow of activities within a system. It illustrates the sequence of actions, decision points, and parallel processes involved in completing a specific task or use case. In the Gokyo Bistro Management System, the Activity Diagram is used to model the flow of operations for a selected use case, such as table booking or order placement, providing a clear visualization of the process from start to end.

The diagram includes elements such as initial nodes, activity states, decision nodes, and final nodes, which together depict the logical flow of control within the system. It helps in understanding how different activities are coordinated and how decisions affect the progression of tasks. By representing the workflow in a structured manner, the Activity Diagram enhances clarity, improves communication among stakeholders, and supports efficient system design.

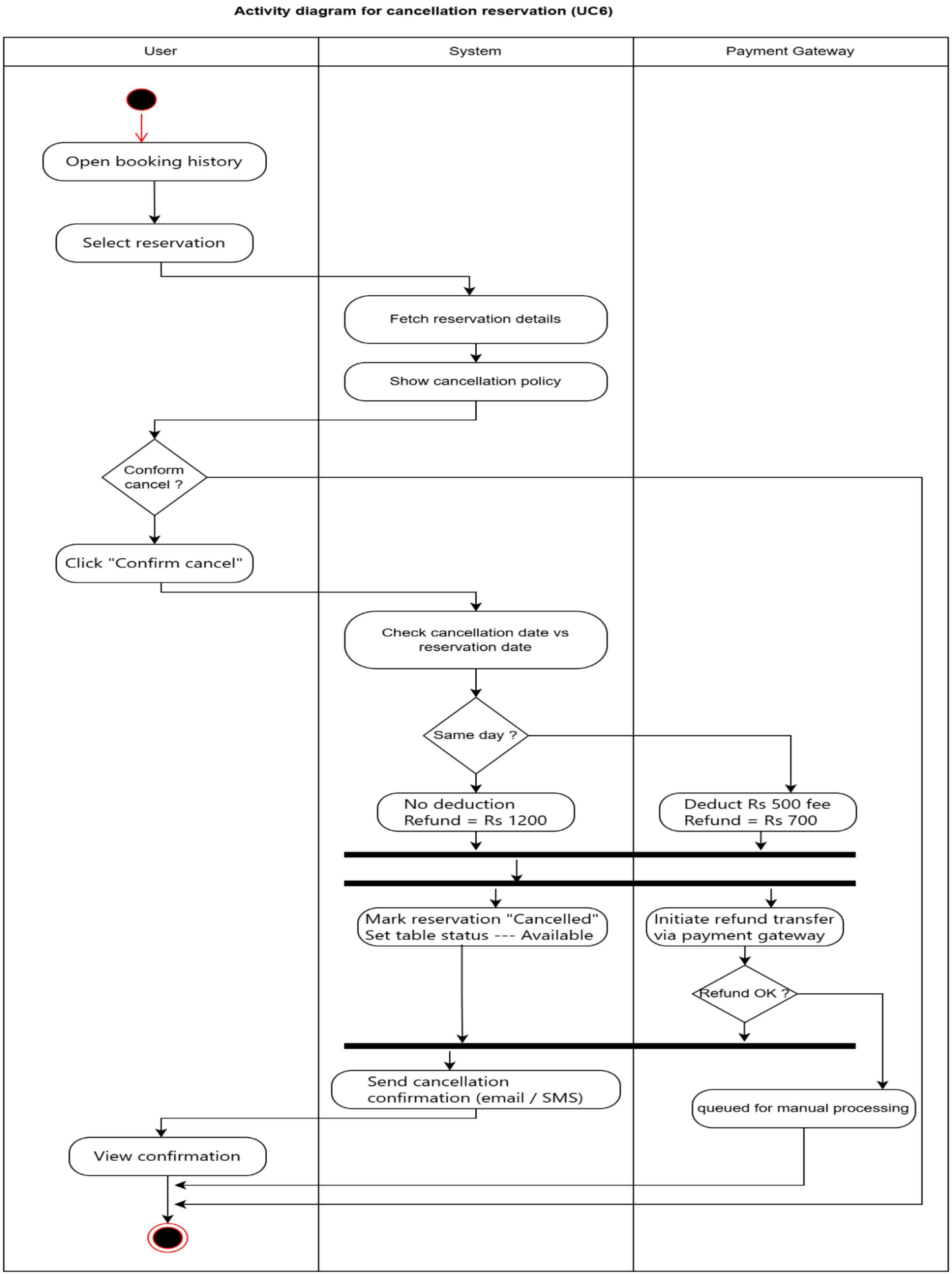

Figure 5: Activity Diagram

8 Class Diagram

The Class Diagram is a fundamental component of Object-Oriented Analysis and Design (OOAD) that represents the static structure of a system. It illustrates the system's classes, their attributes, methods, and the relationships between them. In the Gokyo Bistro Management System, the Analysis Class Diagram is developed to identify and organize all relevant domain classes derived from the use cases, such as User, Admin, Member, Booking, Order, Payment, and Menu.

The diagram also highlights key object-oriented relationships including inheritance, association, aggregation, and composition. Inheritance is used to represent the generalization between classes such as User and its specialized types (Admin, Member, Visitor). Aggregation and composition are used to show whole-part relationships between entities such as orders, bookings, and payments. Additionally, multiplicities are included to indicate how many instances of one class are related to another. This structured representation helps in understanding the system architecture and serves as a foundation for implementation.

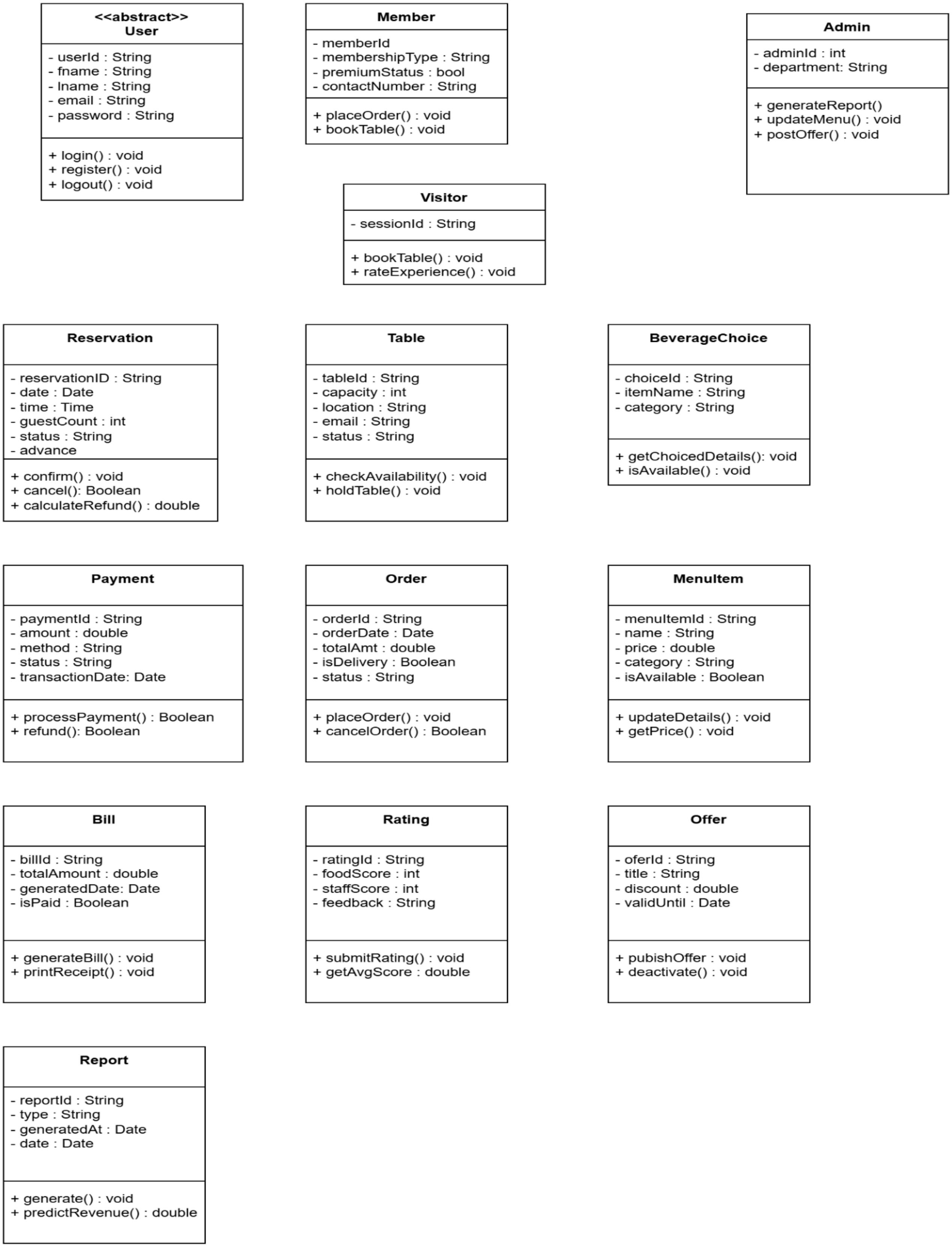

Figure 6: Class Diagram

9 Further Development

After completing the analysis and design phase of the Gokyo Bistro Management System, the next step is to proceed with the system development and implementation. This phase focuses on transforming the designed models into a functional software system. A well-defined development plan is essential to ensure that the system is built efficiently, meets user requirements, and maintains high quality standards. The following sections outline the chosen development methodology, architectural design, design patterns, development approach, testing strategy, and maintenance plan.

9.1 Development Methodology

The Agile (Scrum) methodology is selected for the development of this system due to its flexibility and iterative nature. Agile allows the project to be developed in small increments called sprints, where each sprint delivers a functional part of the system. This approach enables continuous feedback, faster delivery, and easier handling of changing requirements. Regular meetings, such as sprint planning and reviews, ensure effective communication and progress tracking throughout the project.

9.2 Architectural Choice

The system will follow the Model-View-Controller (MVC) architecture. In this architecture:

- The **Model** represents the data and business logic (e.g., Booking, Order, Payment).
- The **View** represents the user interface (e.g., web pages, mobile screens).
- The **Controller** handles user input and controls the interaction between Model and View.

This separation of concerns improves scalability, maintainability, and ease of development.

9.3 Design Patterns

Several design patterns will be implemented to improve system design and reusability:

- **Singleton Pattern**: Used for database connection to ensure a single instance is maintained.
- **Factory Pattern**: Used to create different types of users (Admin, Member, Visitor) dynamically.

- **Observer Pattern** (optional enhancement): Can be used for notifying users about offers or order updates.

These patterns enhance code organization, flexibility, and maintainability.

9.4 Development Plan

The development will be carried out using modern tools and technologies:

Tools and Technologies

- **Frontend**: HTML, CSS, JavaScript (or React)
- **Backend**: Node.js / Java / PHP
- **Database**: MySQL
- **Version Control**: GitHub

Feature Development Priority

1. Login and Registration
2. Book Table
3. Set Beverage choices
4. Payment
5. Generate Report
6. Bill Generation
7. Order placement
8. Update Menu
9. Rate the experience
10. Post offers and schemes

Development will be performed in iterative sprints, with each sprint focusing on specific features.

9.5 Testing Plan

Testing is essential to ensure the reliability and quality of the system. The following testing strategies will be used:

- **Unit Testing**: Testing individual components or modules
- **Integration Testing**: Testing interaction between modules
- **System Testing**: Testing the complete system
- **User Acceptance Testing (UAT)**: Ensuring the system meets user requirements

Testing will be conducted throughout the development lifecycle to identify and fix issues early.

9.6 Maintenance Plan

After deployment, the system will require continuous maintenance to ensure smooth operation. The maintenance plan includes:

- **Corrective Maintenance**: Fixing bugs and errors
- **Adaptive Maintenance**: Updating the system according to new requirements
- **Perfective Maintenance**: Improving system performance and usability

Regular updates and monitoring will ensure that the system remains efficient, secure, and up-to-date.

10 Prototype

Prototyping is an essential step in the software design process that helps visualize the user interface and overall user experience of the system before actual implementation. It provides a preliminary representation of how the system will function and allows stakeholders to understand and evaluate the design. For the Gokyo Bistro Management System, multiple user interface prototypes have been developed to cover the major features of the system, including user management, table booking, ordering, payment, and administrative functionalities.

These prototypes are designed with a focus on usability, simplicity, and efficiency, ensuring that users can easily navigate and interact with the system. Each prototype represents a specific screen or functionality, providing a clear understanding of how different components of the system are interconnected.

1. Login Page

The login page allows registered users such as Admins and Members to access the system by entering their credentials. It includes input fields for username and password along with a login button and error handling for invalid inputs.

2. Registration Page

This page enables new users (visitors) to create an account by providing necessary details such as name, email, contact number, and password. It ensures user data validation before successful registration.

3. Home Page

The home page serves as the main interface of the system, displaying navigation options such as booking tables, ordering food, viewing menu, and accessing offers.

4. Table Booking Page

This page allows users to select a date, time, and number of guests to reserve a table. It also provides options to check availability before confirming the booking.

5. Table Availability Page

This interface displays available and occupied tables for a selected time slot, helping users make informed booking decisions.

6. Beverage Selection Page

Users can optionally select preferred beverages while booking a table. Premium members can choose special drinks such as wine or whiskey.

7. Payment Page

This page facilitates online payment for table reservations and orders. It includes payment options and displays confirmation upon successful transaction.

8. Food Ordering Page

Members can browse menu items, select food and beverages, and add them to their cart for ordering, including home delivery options.

9. Cart Page

The cart page shows selected items, quantity, and total price. Users can modify their order before proceeding to checkout.

10. Bill/Receipt Page

This page generates a detailed bill including ordered items, prices, taxes, and total amount. It also provides an option to download or print the receipt.

11. Admin Dashboard

The admin dashboard provides an overview of system activities, including bookings, orders, revenue, and user management options.

12. Menu Management Page

Admins can add, update, or delete menu items through this interface, ensuring that the menu remains up-to-date.

13. Report Generation Page

This page allows admins to generate financial reports and analyze revenue trends, supporting decision-making and future predictions.

14. Feedback Page

Users can rate their experience based on food quality, service, and ambience, and provide additional comments.

15. Offers and Discounts page

Admins can create and manage promotional offers and discount schemes, which are displayed to users on this page.

11 Conclusion